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ACADEMIC PORTFOLIO

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1. Injection 3D Concrete Printing (I3DCP)



• Ph.D research topic "Injection 3D Concrete Printing – Integration of Structural Design and Automated Robotic Fabrication for Space Truss", supported by Volkswagen Foundation and Matthäi Foundation, Germany;
• AMC - TRR277, A09 subproject, supported by Deutsche Forschungsgemeinschaft (DFG) - German Research Foundation;
• Supervised by Prof. Dr. **Norman Hack** (TU Braunschweig, Germany) and Prof. Dr. **Pierluigi D'Acunto** (TU Munich, Germany);
• Parts of the research results have been exhibited at Time Space Existence: Venice Architecture Biennale 2023 (Venice, Italy) [Video].

1) Structural Design and Optimization

(1) Journal articles



• **Y. Xiao**, N. Hack, H. Kloft, D. Lowke, I. Mai, and P. D'Acunto, "Constraint-Based Form Finding of Space Trusses for Injection 3D Concrete Printing Through Vector-Based Graphic Statics", *Additive Manufacturing*. 104751, ISSN: 2214-8604, DOI: https://doi.org/10.1016/j.addma.2025.104751 (2025).

CiteScore: 19.8 | Impact Factor: 10.3

Keywords: Vector-based Graphic Statics (VGS), Injection 3D Concrete Printing (I3DCP), conceptual design, constraint-based structural optimization, stepwise optimization, combined multi-constraint.

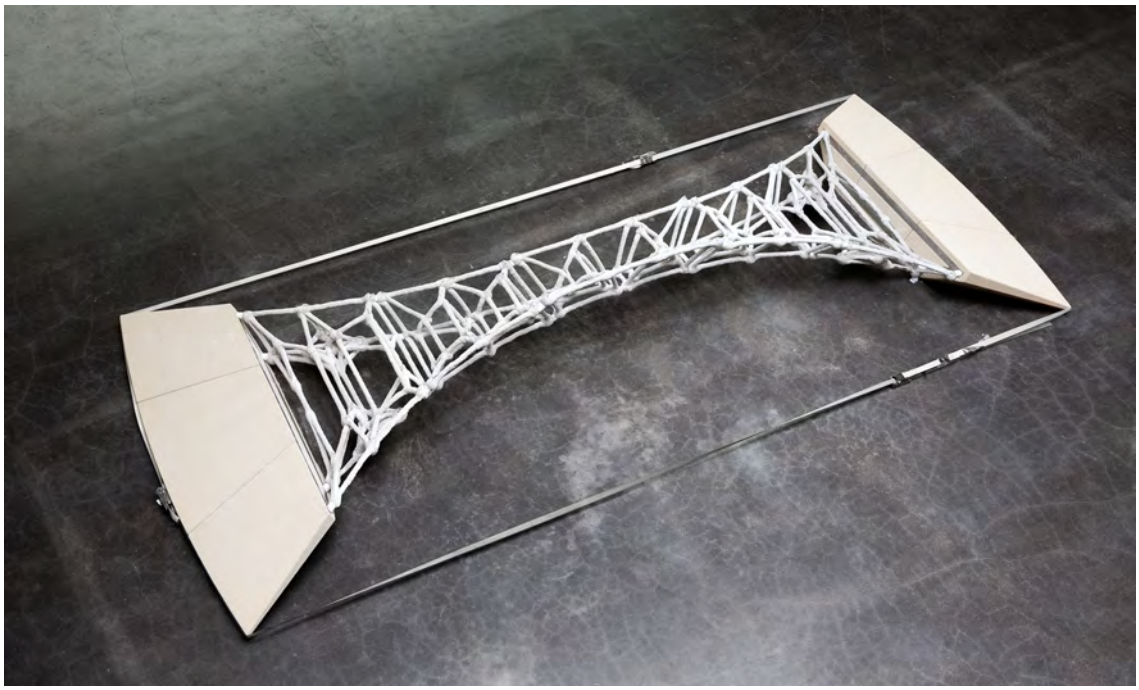


Figure 1..1: I3DCP bridge with 3m span, 0.5m height, and 4.2m total length.

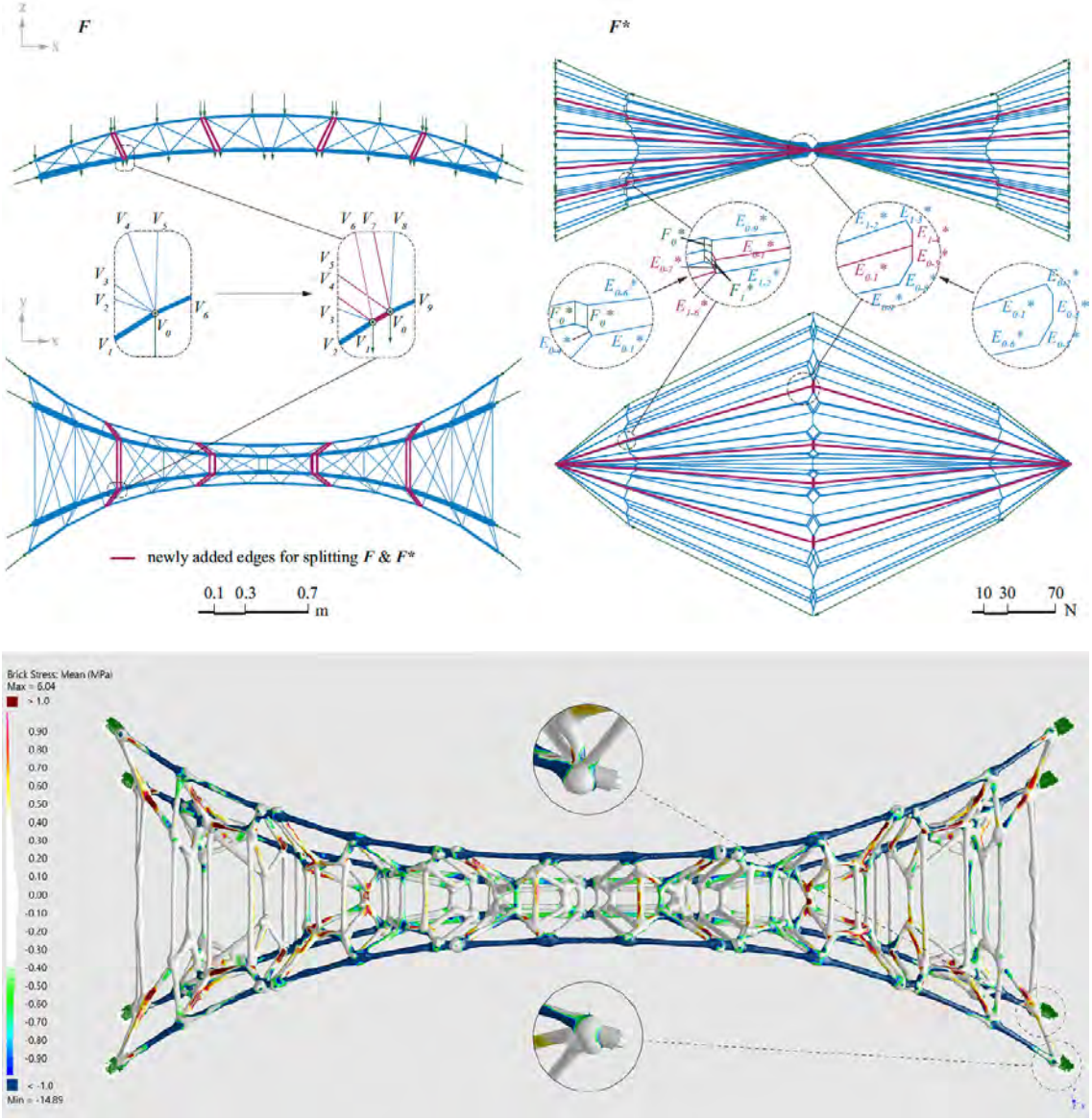


Figure 1..2: Precise segmentation of the I3DCP Bridge by cutting both form and force diagrams along specific edges and nodes. Below is the FEM result of the bridge under 16 × self-weight load.

(2) Conference papers

• **Y. Xiao**, Y. Shen, N. Hack and P. D'Acunto, "A structure- and fabrication-informed strategy for the design of lattice structures with Injection 3D Concrete Printing", in *Proceedings of IASS Annual Symposia* (Beijing, 2022), 1–12. URL.

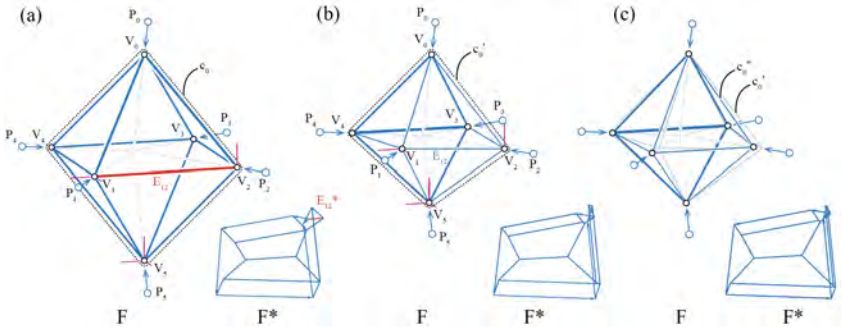


Figure 1..3: Inner force solution for $\boldsymbol{c}_0$ polyhedron.

2) Automated Robotic Assembly Planner

(1) Journal articles



• **Y. Xiao**, A. Vandenberg, D. Lowke, I. Mai, P. D'Acunto, H. Kloft and N. Hack, "Automated Robotic Assembly Planning of Space Trusses for Injection 3D Concrete Printing". *Automation in Construction*. 106440, ISSN: 0926-5805, DOI: https://doi.org/10.1016/j.autcon.2025.106440 [Video] (2025).

CiteScore: 19.2 | Impact Factor: 9.6

Keywords: Injection 3D Concrete Printing (I3DCP), robotic fabrication, robotic assembly planning, sequence and motion planning, constraint satisfaction problem (CSP), space trusses.

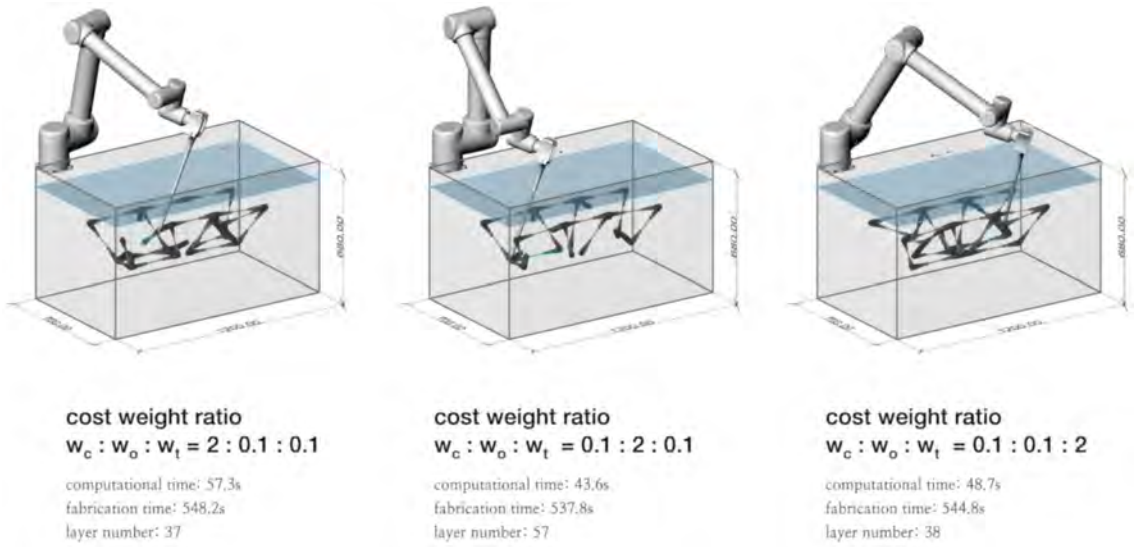


Figure 1..4: The intermediate states of the printing process of one piece of the bridge with different cost weight ratio, resulting in different trajectories.

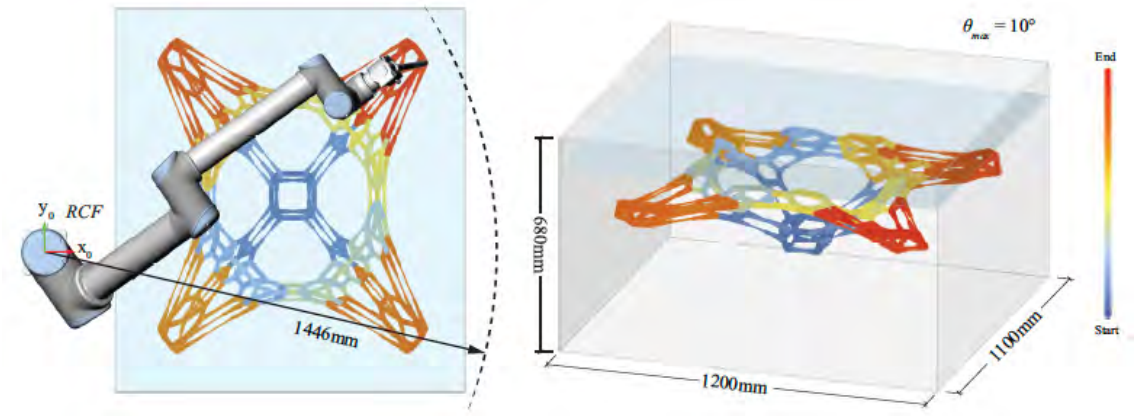


Figure 1..5: Extrusion sequences are color-coded from dark blue to light red.

(2) Conference papers

• **Y. Xiao**, N. Khader, A. Vandenberg, D. Lowke, H. Kloft and N. Hack, "Injection 3D Concrete Printing (I3DCP) Combined with Vector-Based 3D Graphic Statics", in *Third RILEM International Conference on Concrete and Digital Fabrication. DC 2022*. RILEM Bookseries 37 (Springer International Publishing, Cham, 2022), 43–49. isbn: 978-3-031-06116-5. URL.



Figure 1..6: I3DCP table, form and force diagrams.

(3) Magazine articles

• N. Hack, H. Kloft, I. Mai, P. D'Acunto, **Y. Xiao** and D. Lowke, "Injection 3D Concrete Printing: From Structural Geometry to Fabrication". *Architectural Design* 94. URL (2024).

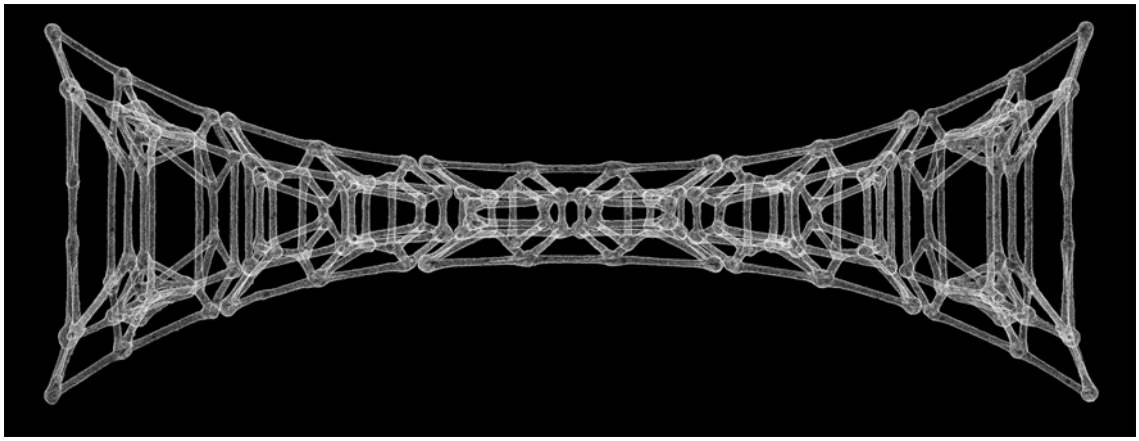


Figure 1..7: Point cloud of the scanned I3DCP bridge.

2. Form-finding Tools

3) Alternating Direction Residual Force Reduction

(1) Journal article



• Y. Shen† and **Y. Xiao**†, "Alternating Direction Residual Force Reduction". *Computer Methods in Applied Mechanics and Engineering.* 119021, ISSN: 0045-7825, DOI: https://doi.org/10.1016/j.cma.2026.119021 (2026).

CiteScore: 13.7 | Impact Factor: 7.6

Keywords: Network structure, initial equilibrium problem, residual reduction , stress state, best-fit inverse form-finding.

Highlights:
(1) A residual-driven equilibrium solver applicable to any user-defined network topology;
(2) An orthogonal alternating search strategy in form and force spaces, offering a methodologically distinct approach to equilibrium computation;
(3) Linear computational complexity with natural support for parallel implementation;
(4) A unified geometric–algebraic framework that accommodates constrained form finding as well as best-fit inverse form-finding tasks.

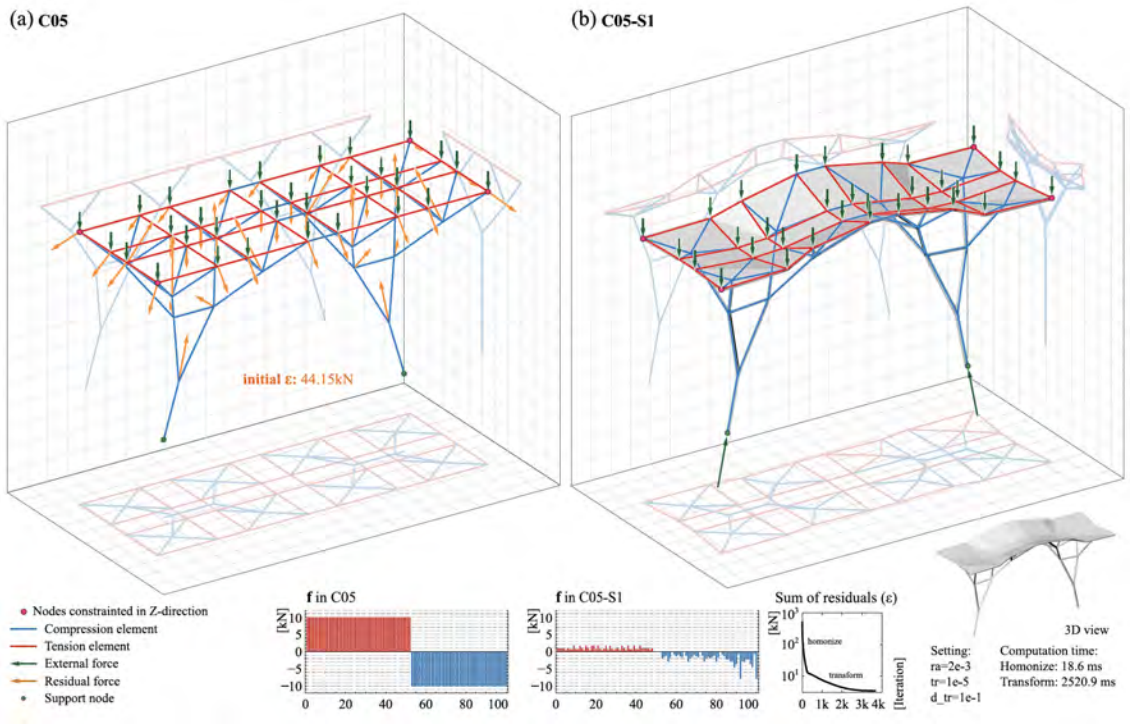


Figure 2..1: ADRFR form-finding of the tree-canopy benchmark.

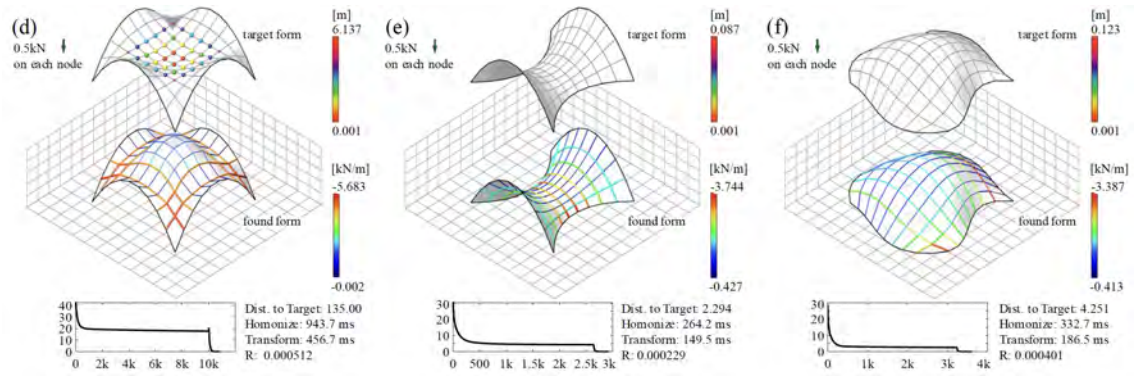


Figure 2..2: ADRFR inverse form-finding results.

4) FDM - Family

(1) GitHub repository (private)

- A collection of different functions of conventional FDM, Non-linear FDM and NFDM (Link: FDM - Family).

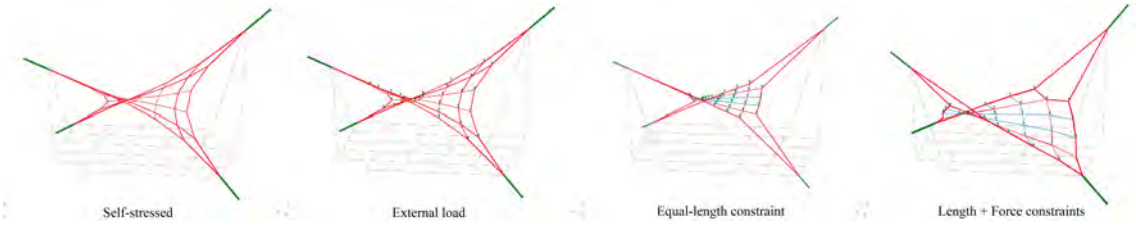


Figure 2..3: Form-finding results using FDM with different constraints.

5) VGS + NLFDM

(1) Conference papers

- Y. Shen†, **Y. Xiao**†, F. He, and P. D'Acunto, "Structural Form-finding Integrating Vector-based Graphic Statics and Non-Linear Force Density Method", in *Proceedings of IASS Annual Symposia* URL (ETH Zurich, 2024)

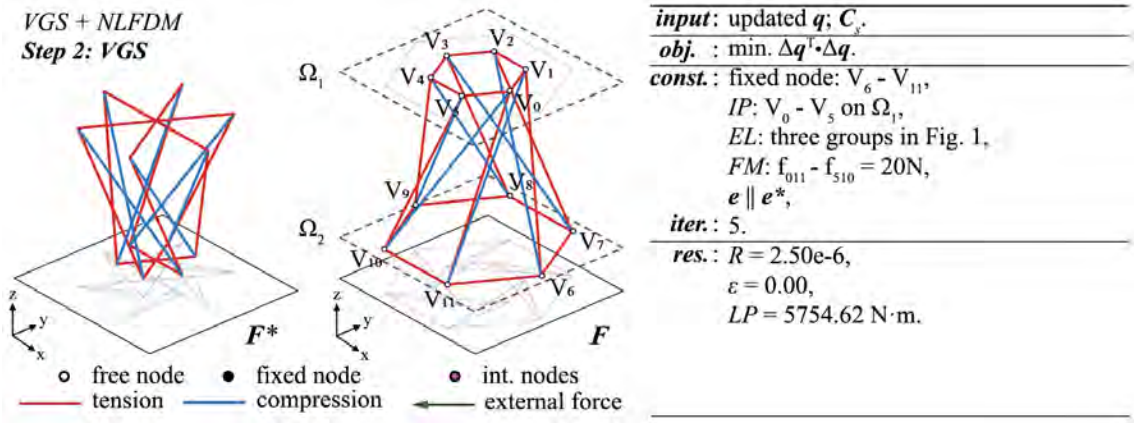


Figure 2..4: Form-finding results of a tensegrity using a combination of NLFDM and VGS.

6) Bending- and Torsion-Active Structures

(1) Master thesis [Video]

- Guided by Prof. Dr. Toni Kotnik
- Aalto University, Finland

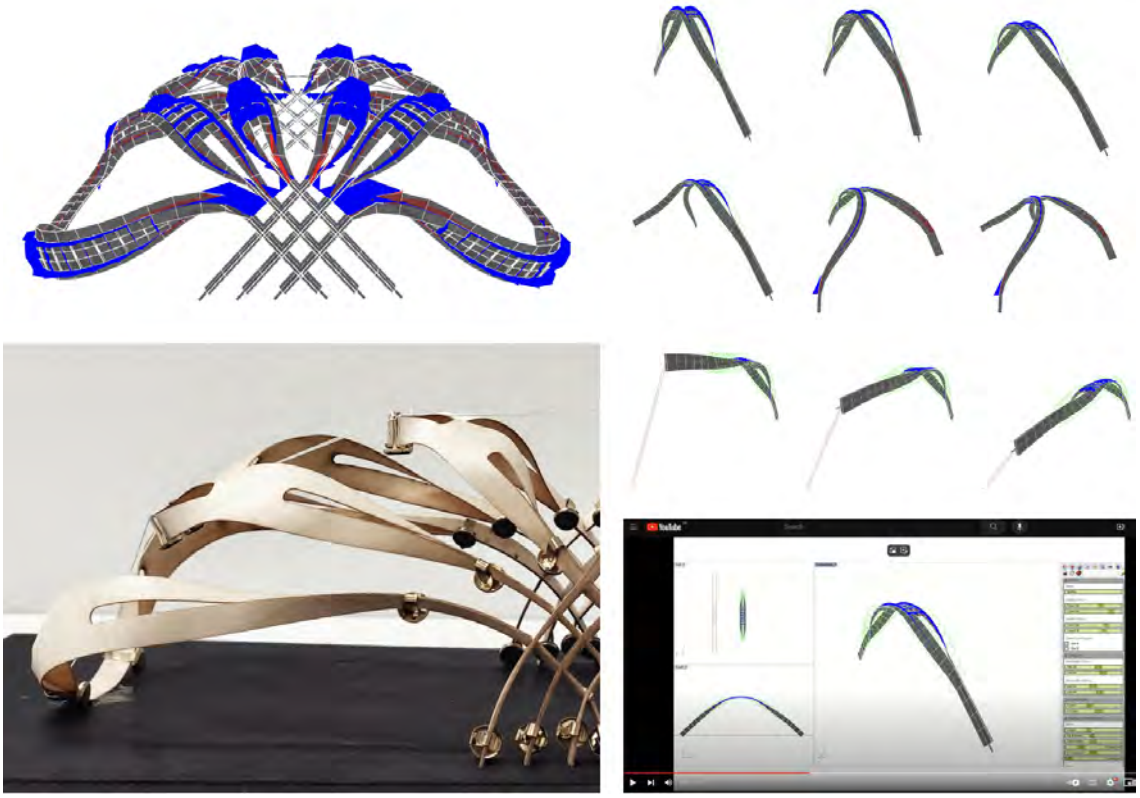


Figure 2..5: Bending- and torsion-active structure prototypes.

3. Digital Twin

7) VGS in Augmented Reality

(1) Conference papers

• N. Hack†, **Y. Xiao**†, C. Jantzen, J. Tesche and K. Mawas, "Design and Fabrication of Structures with Graphic Statics and Augmented Reality", in *Proceedings of IASS Annual Symposia* URL (ETH Zurich, 2024).

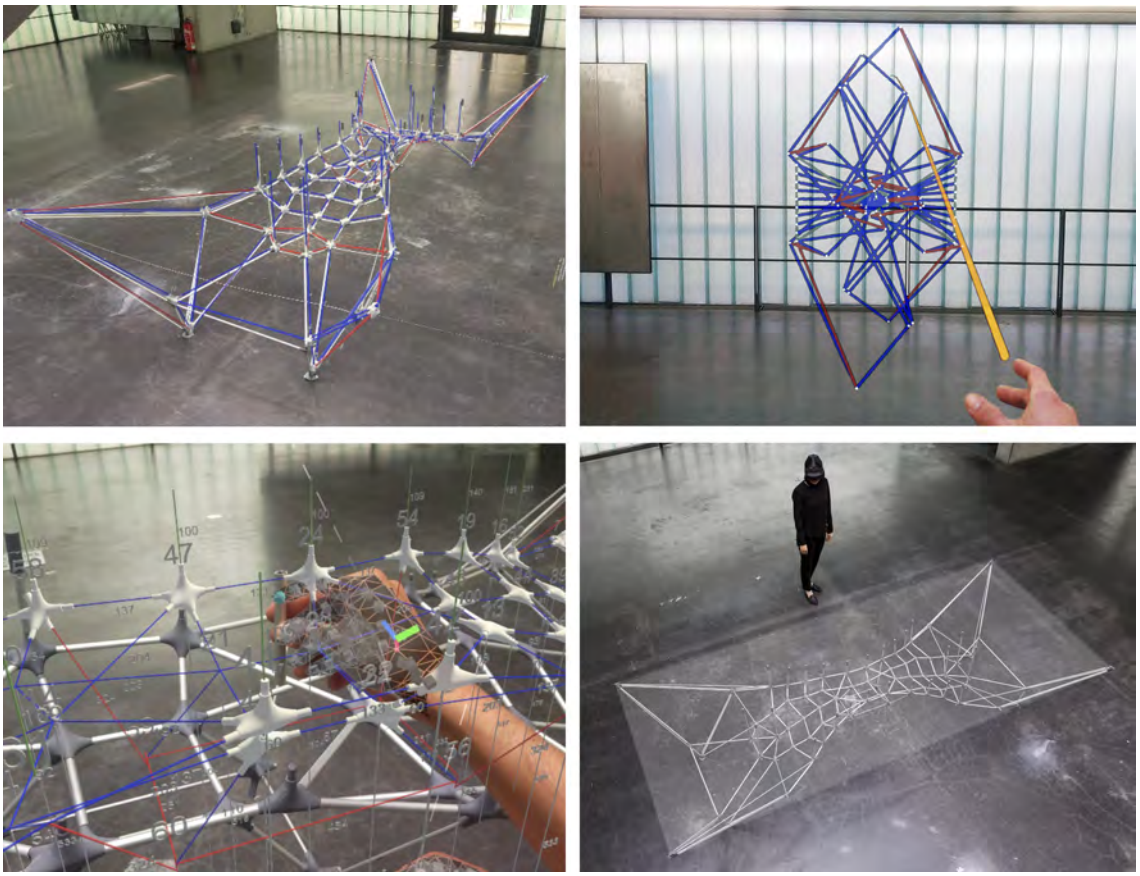


Figure 3..1: AR and VGS enhanced design and fabrication.

4. Teaching Experiences

8) Building Technology 1 (2025-2026), The University of Hong Kong

Period: 12.2025 – 04.2026
Instructors: Zongshuai Wang, **Yinan Xiao**
Professor: Kam-Ming Mark Tam

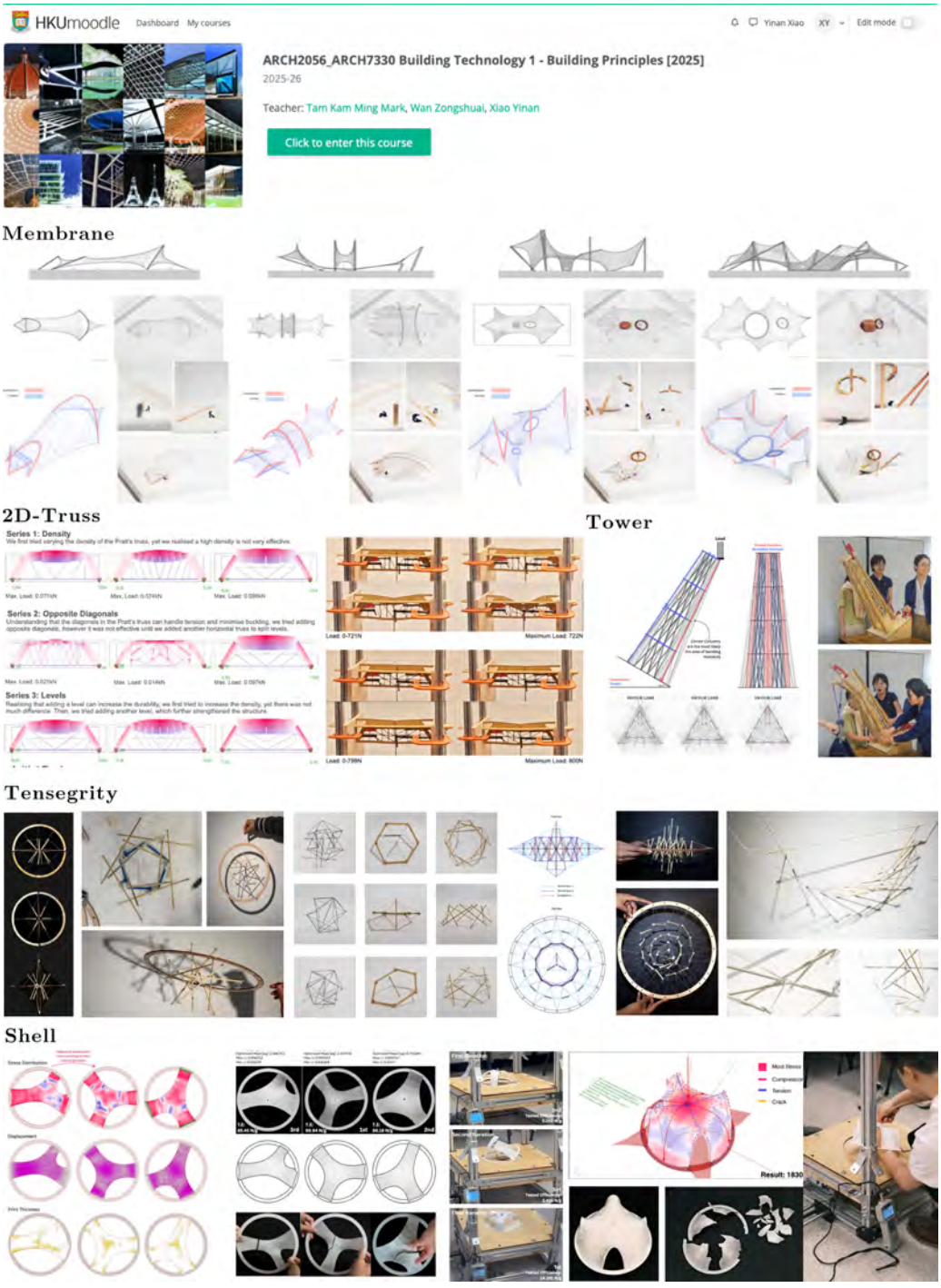


Figure 4..1: Tutor and coding tutorials, helping students to explore different structure kinds, including membrane, 2D truss, tensegrity, high-rise tower, and shell structures.

9) 15th Digital Futures Summer Workshops (2025), Tongji University

Period: 06.2025 – 07.2025
Instructors: **Yinan Xiao**, Feifan He, Yixi Wang, Shizi Ying, and Jie Tang
Professor: Pierluigi D'Acunto, Philip Feng Yuan, Patrick Ole Ohlbrock, Yuchi Shen, and Junwang Yu



Figure 4..2: Instructor, exploring deployable structures with vector-based graphic statics. Result is an umbrella designed and assembled during the workshop.

10) Digital Building Fabrication Design Studio, TU Braunschweig

Period: 08.2021 – 11.2021
Instructors: **Yinan Xiao**, Noor Khader
Professor: Norman Hack, Pierluigi D'Acunto

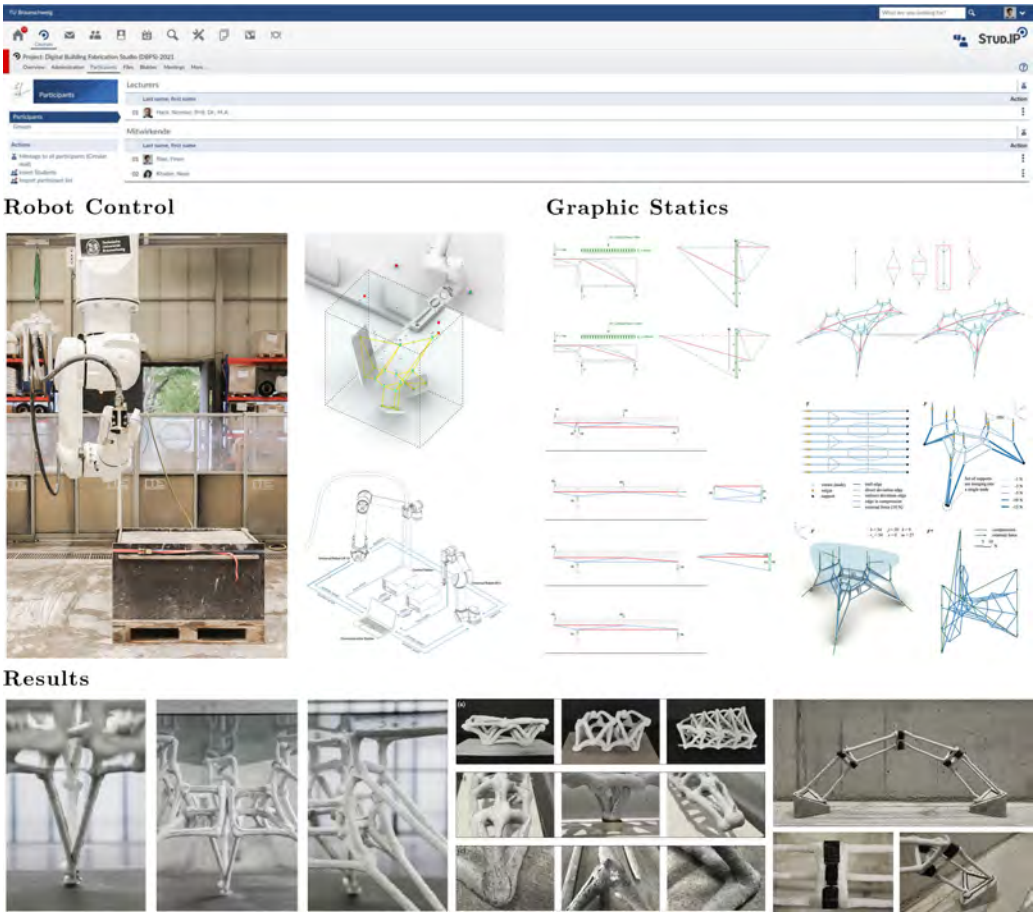


Figure 4..3: Instructor, giving lectures to build up principle knowledge about robot control and graphic statics. The course results are prototypes fabricated by students.

11) LafargeHolcim 3D Printing Housing Hack (semester course), TU Braunschweig

Period: 01.2021 – 04.2021
Instructors: **Yinan Xiao**, Annahita Meshkini, Philipp Rennen, Noor Khader
Professor: Norman Hack

Conference paper: **Y. Xiao**, N. Hack, H. Kloft, "Digital Structural Design And Shotcrete 3D Printing Strategies for Lightweight Reinforced Concrete Beam-Grid Structures", in *Proceedings for the 6th fib International Congress* URL (Oslo, 2022).



Figure 4..4: Instructor, leading two group of students to design and fabricate two demonstrators which won Holcim 3D Printing Hackathon competition 2nd and 4th prize, using shotecrete technology.